

SUMMATIVE ASSESSMENT: EVALUATING LEARNING OUTCOMES IN THE FORM OF TERMINAL/FINAL EXAMINATION

Summative assessment is a fundamental component of the educational process, designed to evaluate student learning at the conclusion of an instructional period. Unlike formative assessment, which provides ongoing feedback during the learning process, summative assessment serves as a final measure of what students have learned. Typically administered at the end of a unit, course, or academic year. Summative assessment aims to determine the extent to which educational objectives have been achieved.

The primary purpose of summative assessment is to assess the overall effectiveness of instruction and learning. It provides a conclusive evaluation of student performance, often in the form of tests, final projects, or standardized exams. These assessments generate grades or scores that reflect a student's achievement in a given subject area over a specific period or time duration.

Summative assessment is often used to make critical decisions regarding student progression, certification, or placement in subsequent educational levels. Additionally, summative assessments provide valuable data that inform curriculum development and instructional strategies. By analyzing summative assessment results, educators can identify trends, strengths, and weaknesses within their instructional approaches, allowing for improvements in future teaching.

In conclusion, summative assessment plays a critical role in the educational process by providing a final evaluation of student learning. While it differs from formative assessment in its focus and application, it is an essential tool for measuring academic achievement. When balanced with formative assessments, summative assessments contribute to a well-rounded and effective approach to evaluating and supporting student learning.

National Curriculum of Pakistan 2022-23
Assessment Framework
COMPUTER SCIENCE Grade-X (SSC-II)
Details of Content Areas/ SLOs

Content Domain/ Area	SLO No./ Description	Form of Assessment	Cognitive Level (Knowledge, Understanding, Application)	Remarks	Number of Periods required (1 period= 40 minutes)
A: Computing Systems	[SLO CS-10-A-01] Students will be able to understand and describe number systems and encoding schemes for data representation in computer systems Students will understand... • What is Machine level representation of data • What are the different numbering systems (decimal, binary, hexadecimal, octal) • How text is represented digitally using common text encoding (ASCII, Unicode) Students will know • Key terms: ASCII, Unicode, binary, signed, and unsigned numbers, bit, byte, negatives in binary, 1's complement, 2's complement, binary arithmetic, overflow and underflow. • How computers use binary arithmetic. Students will be able to • Understand and explain data representation, instruction sets and addressing modes. • Demonstrate how data is encoded using ASCII and Unicode	Summative for Theory	/ Understanding/ Application	Question(s) will be asked in the Annual theory paper as well as	30
	[SLO CS-10-A-02] Students will be able to explain how system software controls the flow of information between hardware components used for input, output, storage, and processing Students will understand that • What is an operating system • What main tasks an operating system performs • How application programs run on top of operating systems • What are the different types of operating systems	Summative for Theory	Knowledge / Understanding	Question(s) will be asked in the Annual theory paper as well as	

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	Students will know • Key terms: System calls, Processes, threads, synchronization, interrupts, system calls scheduling, deadlock, File system interface, Memory system and virtual memory. • The primary resources managed by an operating system • What a process is and the sequence of events executed for a process to complete Students will be able to explain • How programming languages, operating systems, and architectures interact and how to use each effectively. • The services provided by and the design of an operating system. • The structure and organization of the file system. • What a process is and how processes are synchronized and scheduled. • Different approaches to memory management.				
	[SLO CS-10-A-03] Students will identify and learn common software tools such as translators, integrated development environments, online and offline computing platforms, code repositories, etc. Students will understand that • The types of software and how they are different • Where software is hosted and pros and cons of on-premise vs. in cloud • How programming software helps programmers write code that can then get compiled. • Explain the difference between system software, programming software, application software and driver software. • Identify common examples of the different types of applications • Describe uses of common productivity application software • Describe uses of programming software such as language editors, debuggers, compilers, IDEs, source code repositories and build	Summative for Theory	Knowledge / Understanding	Question(s) will be asked in the Annual theory paper	

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	systems (e.g. Eclipse for Java, Coda for Mac, Visual Studio for multiple languages, GitHub for source code) Students will know • Understand uses of different software and tools • Apply common software tools such as translators, integrated development environments, online and offline computing platforms, code repositories etc. • Definitions of key software tools such as translators, integrated development environments, online and offline computing platforms, code repositories etc. • Difference between on premise and hosted application software • What applications can be used without internet (offline) and which require internet to work (online)				
B: Computational Thinking & Algorithms	[SLO CS-10-B-01] Students will identify common algorithms used to develop software, store, search, or sort information Students will understand 1) How to solve the counting problems a) Basics of a counting problem b) Basic counting principles i) Multiplication ii) Addition iii) Permutation iv) Combination v) The pigeonhole principle vi) Inclusion and exclusion principle 2) Properties of Algorithm a) Input b) Output c) Definiteness d) Finiteness e) Effectiveness	Summative for Theory	Knowledge / Understanding/ Application	Question(s) will be asked in the Annual theory paper	20

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	v) Generality 3) Logical reasoning and will be able to solve - Boolean logic - Verbal logical reasoning - Non-verbal logical reasoning 4) Understand data search and sort, and briefly describe standard algorithms on linear arrays such as linear search, binary search, insertion sort, bubble sort etc. Students will be able to - Apply logical reasoning to refine and solve problems - Apply algorithmic thinking to refine and solve problems - Identify when & where to use key search & sort algorithms - Discuss an algorithm to solve a specific problem.				
	[SLO CS-10-B-02] Develop and apply abstractions to create generalized, modular solutions Students will understand - Steps in an algorithm to solve computational problems - Dry running or Trace Table to run algorithm - Identify logical and syntax errors - Abstractions to create generalized, modular solutions Students will be able to - Use algorithmic approach to solve the computational simple problems - Apply abstractions to create generalized, modular solutions - Create and use dry runs/trace tables to follow an algorithm - Identify syntax/logic errors in code and solve logical errors	Summative for Theory	Knowledge / Understanding/ Application	Question(s) will be asked in the Annual theory	
C: Programming Fundamentals	[SLO CS-10-C-01] Students should be able to differentiate between front-end development, and back-end development of a website Students will understand... Back-end development allows writing code that emits HTML/CSS/JavaScript	Summative for Theory	Knowledge / Understanding	Question(s) will be asked in the Annual theory paper	40

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	- Front-end development deals with HTML/CSS/JavaScript in the browser Students will be able to... - Differentiate between back-end and front-end development	✓			
	[SLO CS-10-C-02] Students should be able to use more advanced HTML/CSS features in an appropriate environment Students will understand... - How HTML tags can be used to show tabular data - How HTML can be used to retrieve inputs from users - How to apply animation movements to HTML components Students will be able to... - Create forms in HTML using an IDE like Visual Studio, Netbeans etc. - Create tables in HTML using an IDE like Visual Studio, Netbeans etc. - Create animations in CSS using an IDE like Visual Studio, Netbeans etc.	Summative for Practical Based Assessment	Understanding/ Application	Lab work will be assessed in the Practical Based Assessment	
	[SLO CS-10-C-03] Students should be able to use more advanced programming constructs (lists, etc.) to create dynamic websites using JavaScript as backend scripting Students will understand... - How to create and use arrays in JavaScript - How to create and use bullet points in HTML Students will be able to... - Create bullet points in HTML that are generated from an array in JavaScript	Summative for Practical Based Assessment	Understanding/ Application	Lab work will be assessed in the Practical Based Assessment	
	[SLO CS-10-C-04] Students should be able to implement complex algorithms that use more complex data structures (lists, etc.) in JavaScript Students will understand... - The array data structure is similar to a list - Finding an element in a list requires iterating through entire list till	Summative for Practical Based Assessment	Understanding/ Application	Lab work will be assessed in the Practical Based Assessment	

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	the element is found Students will be able to... - Write an algorithm that finds an element in a list - Implement an algorithm that finds an element in a list using JavaScript				
	[SLO CS-10-C-05] Students will determine more advanced techniques (unit tests, breakpoints, watches) for testing and debugging their code in JavaScript Students will understand... - The purpose of a unit test - Debugging allows them to analyze code as it runs Students will be able to... - Write simple unit tests for the functions in their code - Set a breakpoint and use it to analyze intermediate values of variables in JavaScript	Summative for Practical Based Assessment	Understanding/ Application	Lab work will be assessed in the Practical Based Assessment	
Domain D: Data and Analysis	[SLO CS-10-D-01] Students will understand and explain the scope of data science, Artificial Intelligence (AI), and Machine Learning (ML), including types of supervised and unsupervised learning models, and their applications to common real-world problems. Students will know 1) Definitions of data science, artificial intelligence and machine learning. They will also learn to differentiate between the three fields and overlapping areas. - Machine Learning skills include - Technique used by AI - Enables systems / machines to learn from large data - Uses supervised and unsupervised learning - Automation and scalability - Uses in fraud detection. - Artificial intelligence skills include: Teach machines to emulate human behavior e.g. natural language	Summative for Theory	Knowledge / Understanding	Question(s) will be asked in the Annual theory paper	20

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	processing, chatbots, image recognition 2) Data science lifecycle: Collecting, processing, analyzing, visualizing and understanding model outcomes (predictions / descriptive). Students will know - The difference between data science, AI and machine learning. Recognize different types of supervised learning vs unsupervised learning models including algorithms. - How to interpret the results from churn prediction and behavioral segmentation models Students will - Differentiate between the three fields of data science, AI, and ML and overlapping areas. - Interpret the results from Churn Prediction and behavioral segmentation through simple examples. - Decide if a problem and its solution is using supervised, unsupervised, or reinforcement learning.				
	[SLO CS-10-D-02] Students will understand and explain the types, uses, and methods of data visualizations and understand the benefits of visualizing data. Students will understand that data and data products (charts, graphs, and statistics) can be analyzed and evaluated, similar to analyzing arguments. Students will - Understand the benefits of visualizing data and appropriate methods to create visualizations. - Know visualization techniques such as measures of center and spread, boxplots, bar plots, histograms, scatterplots, graphical summaries of multivariate data, side-by-side bar plots and association, scatterplots - Understand the connection of databases to machine learning. - Explain sorting, visualizing, and using data. - Read plots (identify the name of the plot, interpret the axes, look	Summative for Theory	Knowledge / Understanding/ Application	Question(s) will be asked in the Annual theory paper	

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	for trends, identify confounding factors). - Understand relational databases (entities, attributes, and keys) Students will be able to... - Develop relational schema representing simple Entity-Relationship diagrams - Create tables, forms, reports, and queries using common operations like SELECT, CREATE, JOIN etc. in any database tool (MS Excel, MS Access, MySQL etc.) - Analyze tables in any database tool and develop simple insights (i.e. sales are increasing/decreasing over time etc.) - Develop E-R data models, physical and logical design. - Represent data with plots on the real number line (dotplots, histograms, bar plots, and boxplots). - Interpret differences in shape, center, and spread in the context of the data sets, accounting for possible effects of extreme data points (outliers).				
	[SLO CS-10-D-03] Students will be able to apply stages of the data science life cycle e.g. understanding a real-world business problem, data gathering, building model, and interpreting results). Students will understand... - Define data science life cycle - Explain different steps of data gathering including the source of data, query the data on a database tool (e.g. MS Access, SQL etc.), collect & store the data, format the data to make it ready for modeling. - Different types of real-world business problems and how to formulate them into a data science problem e.g. define a problem e.g. can you help calculate sales for next year. This becomes a sales predictive model. - The data gathering step e.g. from retail stores asking for the last	Summative for Theory	Knowledge / Understanding/ Application	Question(s) will be asked in the Annual theory paper	

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	three years of sales data. - What kind of algorithms are possible for example simple linear regression or decision tree? - Key performance metrics (for example mean squared error) - Students will be able to formulate a business problem into a data science problem and link to types of problems (e.g. questions “my customers are leaving me” is a churn prediction model)				
E: Applications of Computer Science	[SLO CS-10-E-01] Students will be able to describe uses and applications that are enabled by technologies like IoT, and Blockchain Students will know... - IoT and it's foundational components - Applications that are enabled by IoT - Applications that are enabled by blockchain - The benefits of cloud computing Students will be able to... - Analyze different technologies that use blockchains - Evaluate if a scenario fits an application of IoTs - Evaluate how we can use cloud computing	Summative for Theory	Knowledge / Understanding	Question(s) will be asked in the Annual theory paper	20
	[SLO CS-10-E-02] Students will be able to explain that AI can be applied to specific applications in areas like NLP, Robotics, Speech Recognition, etc. Students will understand... - Specific applications of speech recognition like personal assistants, Quran Memorization applications, Speech-to-text typing applications, Speech recognition for authentication, Speech recognition for surveillance and national security, etc. Students will be able to... - Enumerate and explain the use of AI techniques in different real-world applications like personal assistants, Quran Memorization applications, Speech-to-text typing applications, Speech	Summative for Theory	Knowledge / Understanding	Question(s) will be asked in the Annual theory paper	

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	recognition for authentication, - Specific applications of NLP like email filtering to protect against spam and scams, Language translation, document analysis, predictive text, Sentiment analysis, etc. - Applications of Robotics in rescue and search operations, industrial robotics for manufacturing, vacuum cleaners like Romba robotic operations, farming, etc. - Speech recognition for surveillance and national security, etc.				
	[SLO CS-10-E-03] Students will be able to demonstrate the social implications of AI Students will understand... - That improper use of AI tools can result in injustice to specific groups of people. - AI designers have a responsibility towards ensuring that their algorithms target human benefit Students will be able to... - Show that there are instances where use of AI causes social injustices. - Identify the ethical norms for creating AI algorithms	Summative for Theory	Knowledge / Understanding	Question(s) will be asked in the Annual theory paper	
F: Impacts of Computing	[SLO CS-10-F-01] Understand and apply safe & responsible use of the internet to prevent addiction, promote information and data security Students will understand... - The concerns of technology addiction - The need for cyber security to protect data - The ways in which the internet can be used to promote information for good and bad purposes - The effects of threats to individual privacy and security of data from spam, spyware, cookies, etc. - Basics of security & privacy of cloud computing Students will be able to... - Analyze the impact of new technology laws on digital privacy and	Summative for Theory	Knowledge / Understanding	Question(s) will be asked in the Annual theory paper	20

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	information security • Explain safety and security concepts and strategies including peer pressure and cyberbullying in social media impacts lives • Discuss the need for cyber security in relation to privacy and data security of information • Compare ways software developers protect devices and information from unauthorized access				
	[SLO CS-10-F-02] Evaluate the impact of and apply strategies to prevent cyberbullying/harassment Students will understand... • Basics of data, network and cyber security: backups, access, network monitoring, public and private clouds • Use of hardware and software methods to protect devices Students will be able to... • Discuss ways in which to deal with cyberbullying as a victim or someone you know • Discuss cybercrimes laws, instances of cybercrimes, and common methods of reporting cybercrimes	Summative for Theory	Knowledge / Understanding	Question(s) will be asked in the Annual theory paper	
	[SLO CS-10-F-03] Analyze the impacts of the digital divide on access to critical information Students will understand... • Distribution of computing resources affects the way solutions are designed • Inequitable Access to information impacts human lives Students will know: • the meaning of the digital divide Students will be able to: • Design computing applications that can take into consideration accessibility of information • Strategize to alleviate the impacts of unequal access of information in creating computing solutions	Summative for Theory	Knowledge / Understanding	Question(s) will be asked in the Annual theory paper	

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	Discuss what role students can play to minimize the digital divide				
G: Digital Literacy	[SLO CS-10-G-01] Communicate and publish key ideas and details to a variety of audiences using appropriate digital tools and media-rich resources Students will understand... - Understand how to get information on simple topics and extract key ideas, and present them in different formats - How to identify key ideas and create appropriate copy and graphics to present an idea (e.g. communicating event information in a poster, or communicating product features in a billboard) - Appropriate use cases of social media when communicating key ideas to various audiences (e.g. using YouTube to communicate an opinion via a podcast or using Instagram to show images related to a place or a product, or Facebook to share information with a community etc.) - Best practices in reaching audiences on various digital platforms Students will be able to... - Get information on simple topics, extract key ideas, and create appropriate copy & graphics to present an idea to various audiences	Summative for Theory	Knowledge / Understanding	Question(s) will be asked in the Annual theory paper	05

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	in different formats across various social media platforms (e.g. YouTube, Facebook, Instagram, etc.)				
H: Entrepreneurship in the digital age	[SLO EN-10-H-01] Students will use digital tools to conduct research to collect market and customer insights for a business idea. Students will understand... - Quantitative and qualitative research methods and when to apply them. - How to design questions and collect results for various methodologies - How to interpret research results, particularly in regards to bias in sample selection and correlation vs. causation	Summative for Theory	Knowledge / Understanding	Question(s) will be asked in the Annual theory paper	10
	[SLO EN-10-H-02] Students will pitch a business idea. Students will understand... - How to use research to create customer profiles - Components of a successful elevator pitch - Define effective communication skills required to best articulate a business idea - Difference between a business plan and a pitch document	Summative for Practical Based Assessment	Application	Question(s) will be asked in the Annual PBA	